TASK SPECIFICATION: P6-04 PHASE P6 — ACADEMIC CAMPUS SERVICES & RESOURCE PODS

Faculty Office Hours & Dean's Hub API

Git Branch: `feature/P6-04-office-hours-api` · Target Pull Request: feature/P6-04-office-hours-api

Core Objective & Technical Scope:
College leadership and faculty availability windows management with collision-free student appointment booking.

Priority Level	High	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P6-04-office-hours-api`
Upstream Dependencies	P2-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V9__office_hours.sql migration with btree_gist exclusion on faculty appointment intervals.
2. Implement FacultyWindow and Appointment domain entities.
3. Implement OfficeHoursService managing recurring windows and appointment bookings.
4. Create OfficeHoursController with topic classification (Senior Project, Logic Design, Advising).

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V9__office_hours.sql`
- `officehours/**`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Overlapping appointments physically impossible under database exclusion constraints.
- Faculty members can configure and manage only their own office hours.

VERIFICATION & TEST COVERAGE STRATEGY

- OfficeHoursServiceTest
- AppointmentExclusionConstraintTest
- OfficeHoursControllerTest